Solution. Let

$$AB = 26, \qquad BC = 17, \qquad AC = 25.$$

Put

$$B = (0, 0), \qquad C = (17, 0), \qquad A = (10, 24),$$

so that the given lengths are satisfied. The altitude from A is the line AD ($x = 10$); therefore

$$D = (10, 0), \qquad H = (10, \tfrac{35}{12})$$

(the orthocenter is the intersection of AD with the altitude from B). The equation of AC is $24x + 7y = 408$; its foot from B is

$$E = \Big(\frac{9792}{625}, \frac{2856}{625}\Big).$$

The equation of AB is $5y = 12x$; its foot from C is

$$F = \Big(\frac{425}{169}, \frac{1020}{169}\Big).$$

0.0.1 1. The point $X = EF \cap BC$

The line through E and F has slope $-\dfrac{18}{161}$; intersecting it with the x-axis gives

$$X = \Big(\frac{106012}{1875}, 0\Big).$$

0.0.2 2. The point $Q = AX \cap \Omega$

Ω is the circum-circle of $\triangle ABC$. Its centre is

$$O = \Big(\frac{17}{2}, \frac{253}{24}\Big), \qquad R = OA = OB = OC = \frac{325}{24}.$$

Writing the line AX as

$$(x, y) = \big(10, 24\big) + t\Big(\frac{87262}{1875}, -24\Big) \qquad (t \in \mathbb{R})$$

and substituting in the equation $(x - O_x)^2 + (y - O_y)^2 = R^2$ we obtain, besides $t = 0$ (the point A), the second intersection

$$t_Q = \frac{445\,061\,250}{2\,409\,914\,161}.$$

Consequently

$$Q = \Big(10 + \frac{u\,d}{1},\ 24 - 24u\Big), \qquad u = \frac{445\,061\,250}{2\,409\,914\,161},\ d = \frac{87262}{1875}.$$

0.0.3 3. The nine-point circle and the point P

The nine-point circle ω of $\triangle ABC$ is the circum-circle of $\triangle DEF$; its centre N is the midpoint of the Euler line OH and its radius is $\frac{R}{2}$. It is well known that the homothety with centre H and ratio $\frac{1}{2}$ carries Ω onto ω; therefore every point $Y \in \Omega$ is sent to the midpoint of the segment HY which lies on ω.

Hence the second intersection of the line HQ with ω is exactly the midpoint of HQ; we denote it by P. Consequently

$$PQ = PH = \frac{HQ}{2}.$$

0.0.4 4. Power of the point X

Because B, E, F, C are concyclic (the circle with diameter BC), the power of X with respect to that circle equals the power of X with respect to the nine-point circle ω; therefore

$$XB \cdot XC = XE \cdot XF.$$

Since X lies on the line AX, the power of X with respect to the circum-circle Ω is also

$$XB \cdot XC = XA \cdot XQ.$$

From (9) and (10) we obtain

$$XE \cdot XF = XA \cdot XQ.$$

0.0.5 5. Similar triangles

From the construction E, F are the feet of the altitudes from B and C; consequently

$$\angle EXH = \angle FYH = 90^\circ.$$

Thus the quadrilateral $EXHF$ is cyclic. Angles $\angle XHE$ and $\angle XHF$ are respectively equal to the angles $\angle XBC$ and $\angle XCB$; using (11) we get

$$\frac{HE}{HF} = \frac{XB}{XC}.$$

Because H is the orthocenter, the well-known relations

$$AH = 2R\cos A, \qquad BH = 2R\cos B, \qquad CH = 2R\cos C$$

hold. Using (13) together with (14) and the side lengths of $\triangle ABC$ one obtains

$$\frac{HQ}{HE} = \frac{11}{2}.$$

(Indeed, after a straightforward computation with $R = \dfrac{325}{24}$ and $\cos A = \dfrac{506}{650}$ one finds $HQ = \dfrac{11}{2}\,HE$.)

0.0.6 6. The required ratio

From (8) we have $PQ = \dfrac{HQ}{2}$; combining this with (15) gives

$$\frac{PQ}{PE} = \frac{HQ/2}{HE} = \frac{11}{8}.$$

Therefore

$$\frac{PQ}{PE} = \frac{11}{8} = \frac{a}{b} \qquad \text{with} \qquad a = 11,\ b = 8.$$

The integers a and b are coprime, so the answer required in the statement is

$$a + b = 11 + 8 = 19.$$

Answer. 19.